Department of Computer Science
Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

SUBJECT NAME :	**COMPUTER ORGANIZATION AND ARCHITECTURE**
SUBJECT CODE :	**SECR 1033**
SEMESTER :	**2 - 2023/24**
LAB TITLE :	**Programming 1: Assembly Language Fundamentals**
INSTRUCTION :	**Student is required to have instructor's signature before proceed from each lab work.**
NAME :	• **NABIL AFLAH BOO BINTI MOHD YOSUF BOO YONG CHONG (A23CS0252)** • **NURUL ADRIANA BINTI KAMAL JEFRI (A23CS0258)**
STUDENT INFO :	**SECTION : 02** **EMAIL ;** • **nabilaflahboo@graduate.utm.my** • **adriana04@graduate.utm.my**
COMMENTS :	

Lab # 1

Simple program to familiarize with Program Code, Rebuild & Start Without Debugging

Execute the programs below:

PART 1:

i. **Part A**: Adding and Subtracting Integers

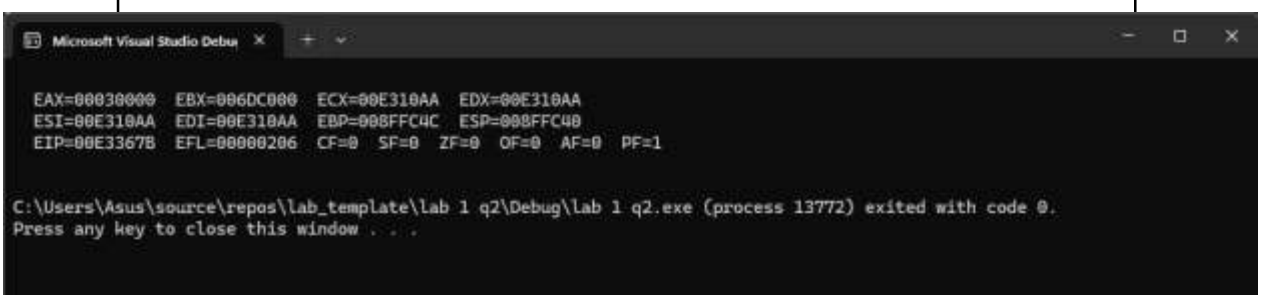
```
TITLE Add and Subtract (AddSub.asm)

; This program adds and subtracts 32-bit integers
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
TOTAL dword 0 ; a variable named TOTAL (declared as DWORD)

.code

main PROC

    mov eax, 123400h ; Set EAX with the value of 123400h
    add eax, 567800h ; Add the content of EAX with 567800h
    sub eax, 77700h ; Subtract content of EAX with 77700h
    mov TOTAL, eax  ; Store content of EAX to TOTAL

    call DumpRegs

    exit
main ENDP

END main
```

Experimental Results:

a) Rebuild & Start Without Debugging

Completed *Fully* *Partially* *: Checked by: ______________*

Screenshot Result:



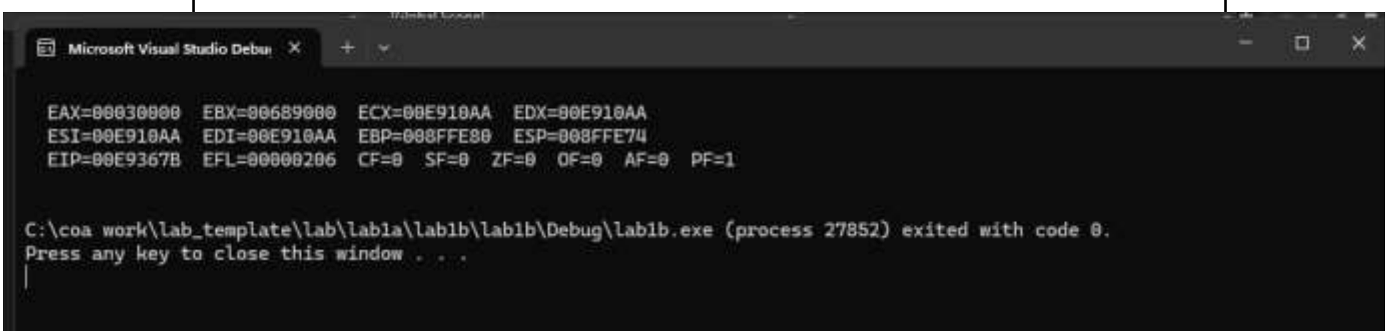
ii. **Part B**: Adding Variables

```
TITLE Add and Subtract, Version 2(AddSub2.asm)

; This program adds and subtracts 32-bit unsigned
; integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
val1  DWORD 10000h
val2  DWORD 40000h
val3 DWORD 20000h
finalVal DWORD ?
.code

main PROC
mov eax,val1      ; start with 10000h
add eax,val2      ; add 40000h
sub eax,val3      ; subtract 20000h
mov finalVal,eax ; store the result (30000h)

call DumpRegs     ; display the registers

exit

main ENDP
END main
```

Experimental Results:
a) Rebuild & Start Without Debugging

Completed   Fully   Partially   : Checked by: ______________

Screenshot Result:



iii. **Part C**: Add and Subtract 8 and 16-Bit Version

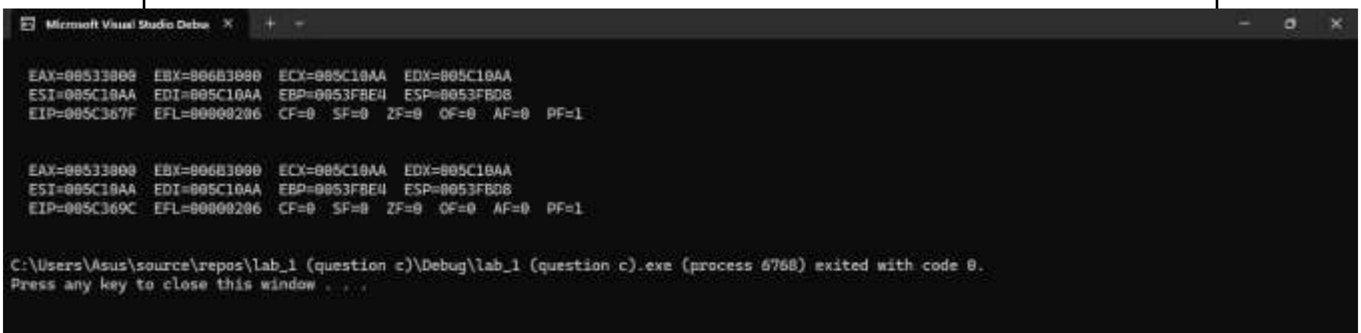
```
TITLE Add and Subtract, Version 3
; This program adds and subtracts 8 and 16 bit
; unsigned integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
valw1 WORD 1000h
valw2 WORD 4000h
valw3 WORD 2000h
finalValw WORD ?

valb1 BYTE 10h
valb2 BYTE 40h
valb3 BYTE 20h
finalValb BYTE ?

.code
main PROC
mov ax,valw1; start with 10000h
add ax,valw2       ; add 40000h
sub ax,valw3       ; subtract 20000h
mov finalValw,ax   ; store the result (30000h)
call DumpRegs      ; display the registers

mov ah,valb1; start with 10000h
add ah,valb2       ; add 40000h
sub ah,valb3       ; subtract 20000h
mov finalValb,ah   ; store the result (30000h)
call DumpRegs      ; display the registers

exit
main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging

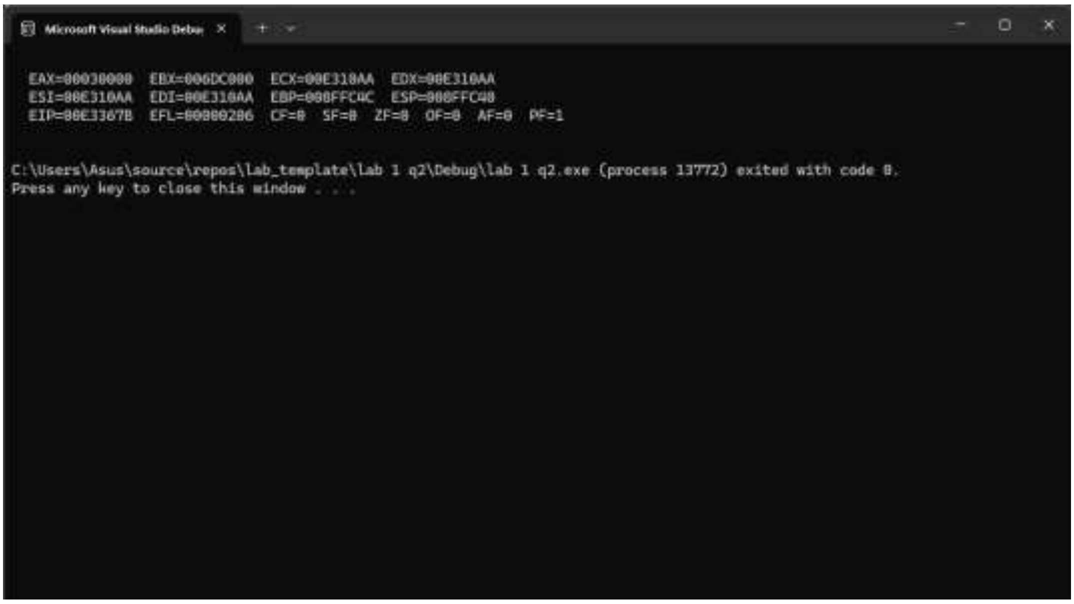
Completed   Fully   Partially   : Checked by: ______________

Screenshot Result:

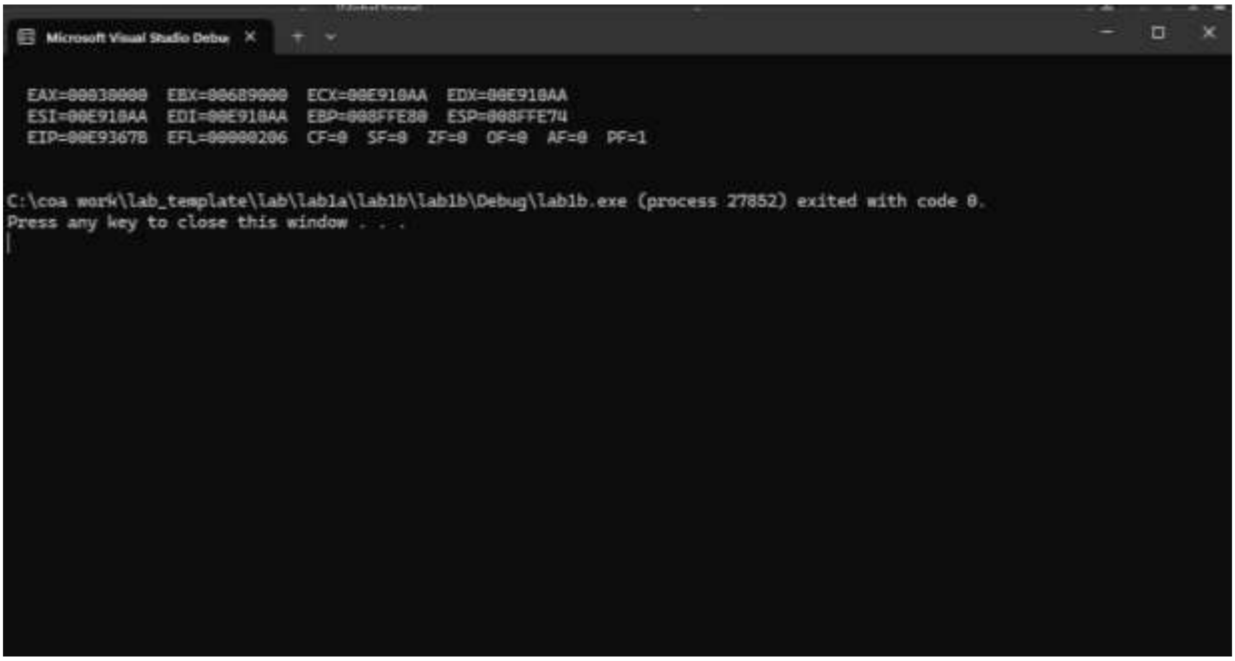


ANSWERS AND SCREENSHOT

PART A:



PART B (WITHOUT DEBUGGING):

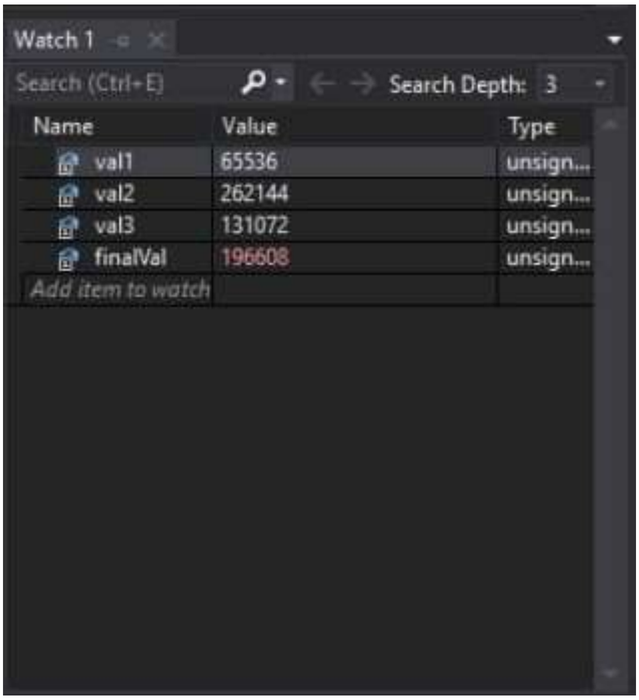


PART B (DEBUGGING):

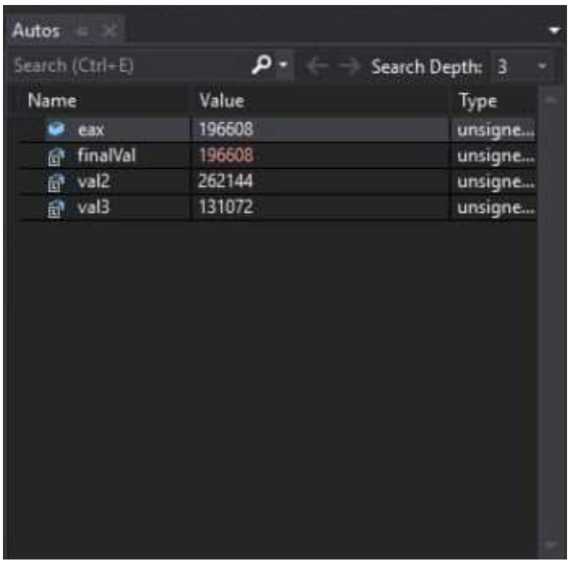
Memoryy

Register:

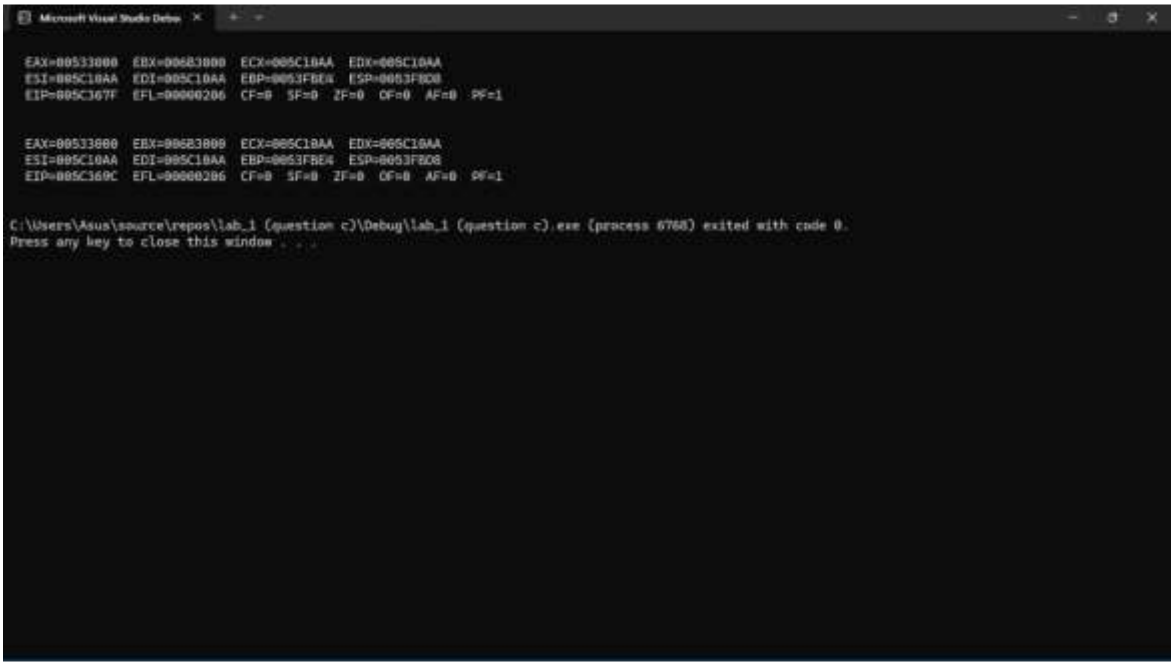
Watch:

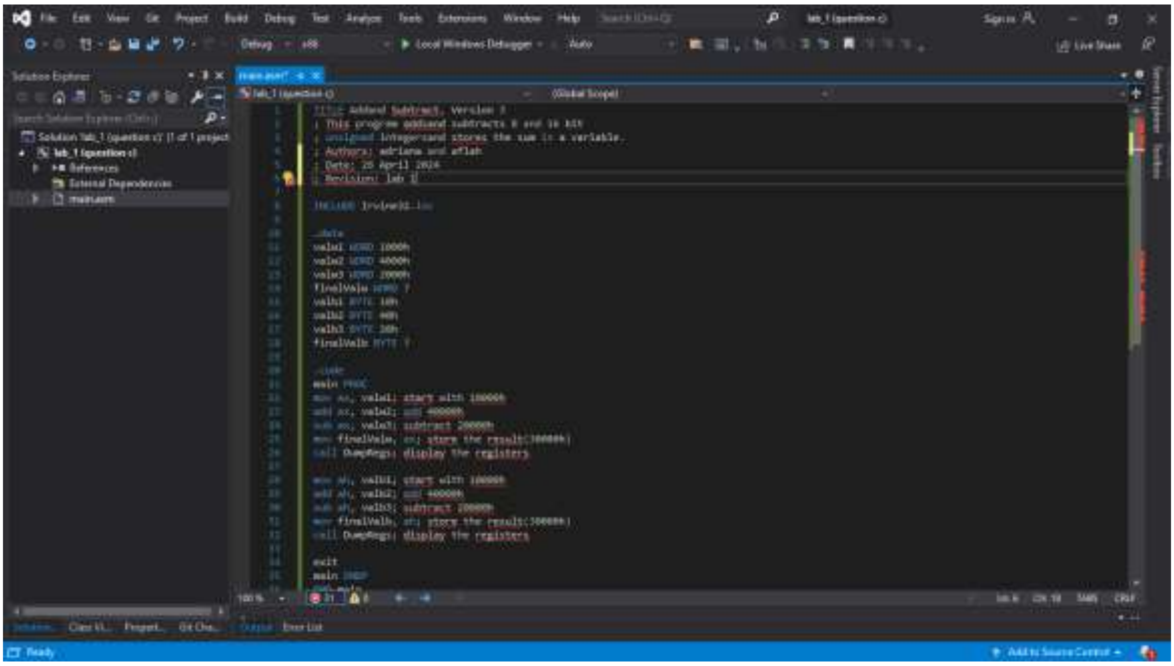


Autos:



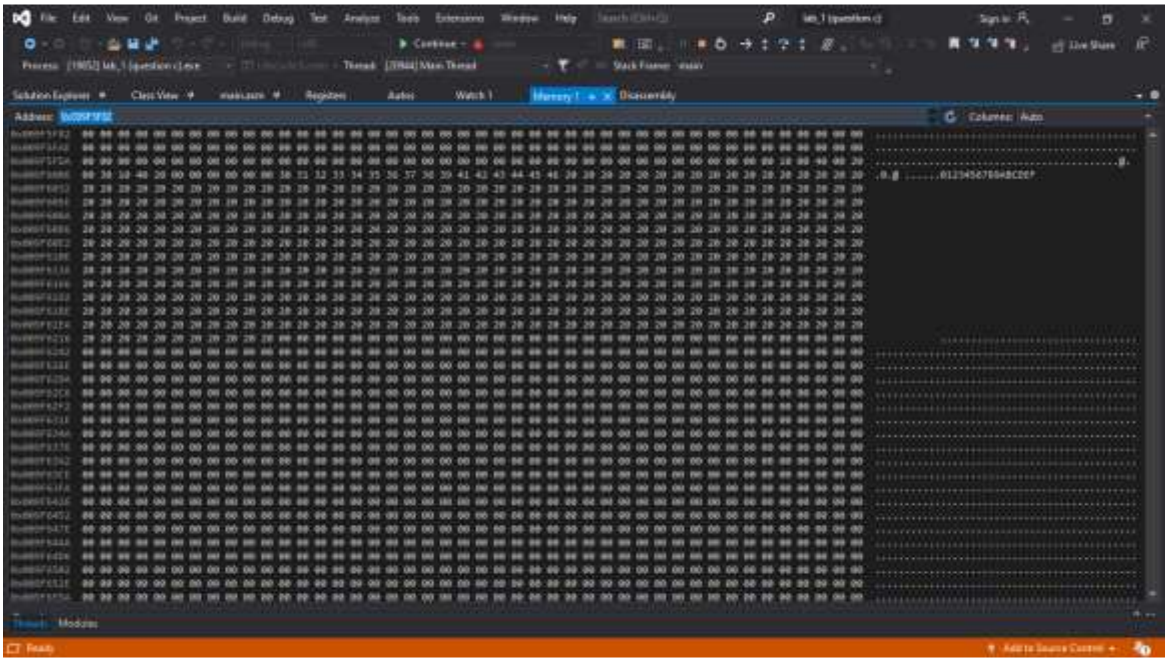
PART C (WITHOUT DEBUGGING):



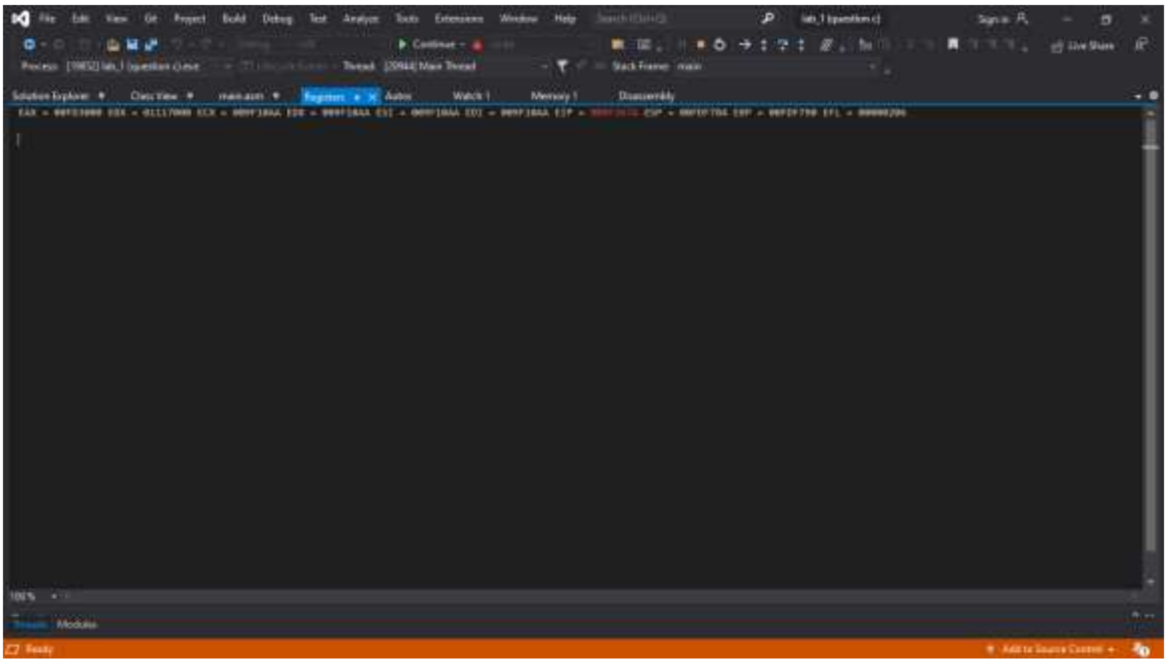


PART C (DEBUGGING):

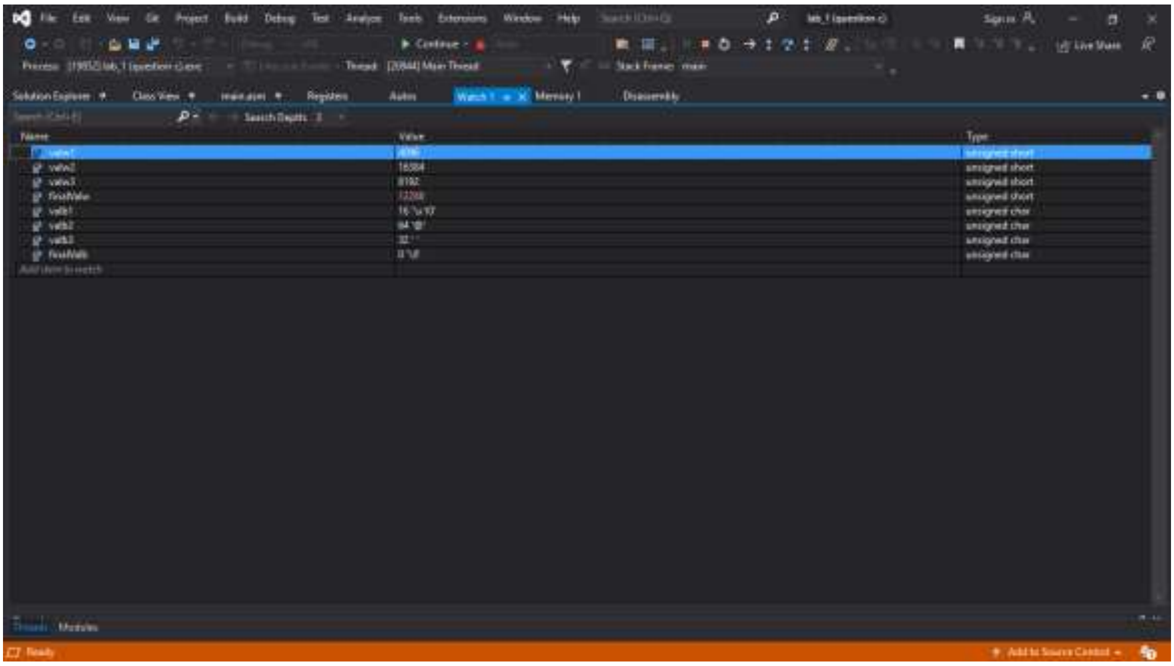
Memoryy



Register:



Watch:



Autos: